VENUGOPAL INDRAJITH HEMASHREE

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EDUCATION

Vellore Institute of Technology, Chennai Aug 2023 – Expected 2027
B.Tech, Computer Science and Engineering (Cyber-Physical Systems) | CGPA: 7.81 / 10

Maharishi International Residential School, Kanchipuram XII: 79% | X: 89%

PROJECTS

TriageIQ — Real-Time LLM Support-Ticket Triage | GitHub 2026
Python, Apache Kafka, Ollama (qwen2.5), Qdrant, FastAPI, SQLite, Docker Compose
- Built an event-driven pipeline that turns incoming support emails (IMAP) into Kafka events, classifies category, priority and sentiment with a local LLM, and drafts replies grounded in similar past tickets retrieved from Qdrant (RAG).
- Kept a human in the loop: a FastAPI + Server-Sent Events agent console streams tickets live, and replies go out over SMTP only after approval; invalid model output is routed to a person instead of being dropped.
- Hardened for real email: Message-ID de-duplication, per-sender rate limiting, auto-reply loop protection and signed sessions with lockout; runs fully locally, so no customer data leaves the machine.

RootLens — Automated Root-Cause Analysis for Microservices | GitHub 2026
Python, FastAPI, Docker Compose, Prometheus, Grafana, SQLite, GitHub Actions
- Built a three-service FastAPI system with fault-injection switches and a Prometheus-based detector that opens an incident after 3 checks at 30%+ failure ratio and resolves it automatically on recovery.
- Wrote an investigator that scores each failing dependency as down, erroring or slow using up-metrics, error ratios and p95 latency, and returns ranked root causes with confidence levels and the downstream services affected.
- Added versioned releases with rollback and a policy-limited remediation controller (dry-run/enforce, action allowlist, rate limits) that rolled back a faulty release automatically in a live test; CI runs unit tests and image builds on every push.

RL-Driven Autonomous Rover for Disaster-Response Reconnaissance 2026
Python, Raspberry Pi 5, BreezySLAM, YOLOv8n / YOLOv9t, A, Reinforcement Learning, Flask*
- Ran LiDAR SLAM (BreezySLAM) on a Raspberry Pi 5 to map the area in real time, overlaid with a risk heatmap of hazard zones from an MQ-2 gas sensor and a camera-based YOLOv9t model.
- Added a risk-score head to YOLOv8n and reweighted its loss toward human detection; a modified A* planner finds the safest exit path, and RL re-plans the sweep around new hazards.

AI Dental Malocclusion Detection with Explainable AI | GitHub Jul 2025 – Sep 2025
Python, PyTorch, YOLOv9, Grad-CAM, OpenCV, Gradio
- Trained a YOLOv9 detector for 31 dental condition classes on panoramic X-rays, reaching 97.1% mAP@50 (96.8% precision, 95.9% recall) on a held-out test set.
- Added Grad-CAM explanations, FDI tooth charting, PDF reports and a Gradio app for real-time use.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS
Backend & Streaming: FastAPI, REST APIs, Server-Sent Events, Apache Kafka, SQLite, MySQL
DevOps & Observability: Docker, Docker Compose, GitHub Actions (CI), Prometheus, Grafana, Git, Linux
AI / ML: LLMs (Ollama), RAG, Qdrant, PyTorch, TensorFlow, YOLOv8/v9, OpenCV, Grad-CAM, MMPose
Embedded / IoT: Raspberry Pi, Arduino, sensor fusion, LiDAR SLAM
Spoken: Tamil (native), English (fluent), Hindi (intermediate), Japanese (basic)

PATENT

IoT Smart Walker with Sensor-Fusion Fall-Risk Prediction — Co-Inventor Filing in progress
Inventive disclosure filed with IIPRD (Khurana & Khurana) | Faculty guide: Dr. Subitha D | Arduino, MPU6050, HX711, HC-SR04
- Designed a fall-risk engine and a Dynamic Stability Reserve metric that fuse bilateral tilt, weight asymmetry and obstacle distance to flag deteriorating stability before a fall, with near-fall detection and a Gait Symmetry Index for rehabilitation.
- Runs fully standalone on an Arduino Uno with no cloud dependency, at an estimated hardware cost under USD 30.

RESEARCH EXPERIENCE

Research Intern — VIT Chennai (Prof. Pavithra) | 1 month May 2025 – Jul 2025
- Ran heatmap-based human pose estimation with MMPose and validated predicted joints against Kinovea motion-analysis measurements.

CERTIFICATIONS & LEADERSHIP

- **IBM** GenAI with watsonx (Jun 2025, certificate) · **NUS** Industry 4.0 (Jun 2025, certificate) · **SuperAI 2025** (Singapore)
- **Team Leader**, CodeYGen Club (coding and design projects) · **Design Lead**, Fitness Club (event creatives)